Problem / Question

Privacy regulations (FERPA/GDPR) restrict the use of predictive AI in K-12 education, preventing early identification of at-risk students. Current centralized models require pooling sensitive data, creating a high security risks and legal barriers for school districts

Engineering Goal

To engineer and validate a scalable machine learning pipeline that identifies students at risk of learning gaps with >85% accuracy, utilizing only de-identified and synthetic data, eliminating the need to transfer sensitive students records.

Variables

Controlled variables (Simulation Parameters)

- Population Size: N = 1000 synthetic students per district node.
- Random Seed: Fixed seed (42) for Numpy to ensure reproducible stochastic generation.
- Feature Scaling: Standard Scaler ($z = (x - \mu) / \sigma$) applied locally within each district silo before training.
- Input Features: A fixed schema of 7 behavioral indicators including time_on_task_min, lms_interactions, and weekend_ratio.

Independent variable (Model Architecture & Input)

- Classifier Architecture: Comparative analysis of 4 distinct model families:
  - Statistical Baseline (Majority Class)
  - Logistic Regression (Linear)
  - Multi-layer Perceptron (Neural Network)
  - Gradient Boosted Decision Trees (XGBoost)

Feature Engineering: Introduction of the Consistency Index (CI) and behavioral interaction terms (late_night_freq × volatility) to measure non-linear risk factors.

Dependent variable (Performance Metrics)

- Predictive Accuracy: Global accuracy on held out test sets.
- Discriminative Power: ROC-AUC Score (Receiver Operating Characteristic - Area Under Curve) to measure the model's ability to distinguish between "At Risk" and "Safe" students without bias.
- Privacy Utility Loss: The performance gap between the centralized XGBoost (~95.8%) and the final Federated Model (~84.33%)

Federated Utility: The accuracy retained by the global model after Federated Averaging (FedAvg) compared to centralized training.

Materials

- Hardware: Macbook Air (M-Series Chip)
- Software Stack: Python 3.9.6, TensorFlow, Scikit-Learn, Numbers
- Libraries: XGBoost, Pandas, NumPy, SHAP (Explainability), Matplotlib
- Data Sources: UCI Student Performance Dataset, Custom Synthetic Data Generator

--- PROJECT ATHENAL: FINAL PRODUCTION VALIDATION ---

Model Architecture	Accuracy	ROC-AUC
Baseline	49.50%	0.500
LogReg	84.17%	0.896
NeuralNet	90.67%	0.964
🏆 XGBoost	95.83%	0.982

✓ PROJECT ATHENAL SCIENTIFIC VALIDATION COMPLETE.

Comparative Analysis of Predictive Performance. Table created by Student Researcher using Visual Studio Code, 2026.

Background Research

The Current Landscape: Centralized EWS

Current Early Warning Systems (EWS) rely on centralized data warehousing. This architecture requires the school districts to transfer sensitive Personally Identifiable Information (PII) to a third party server.

- **The Risk:** Centralization creates a (Single Point of Failure).

One breach at a central repository exposes the records of every student in the system.

- **The Barrier:** Legal and ethical restrictions under FERPA (Family Educational Rights and Privacy Act) and GDPR often prevent districts from sharing data, which leads to Model Isolation where smaller schools lack the data volume to train accurate models.

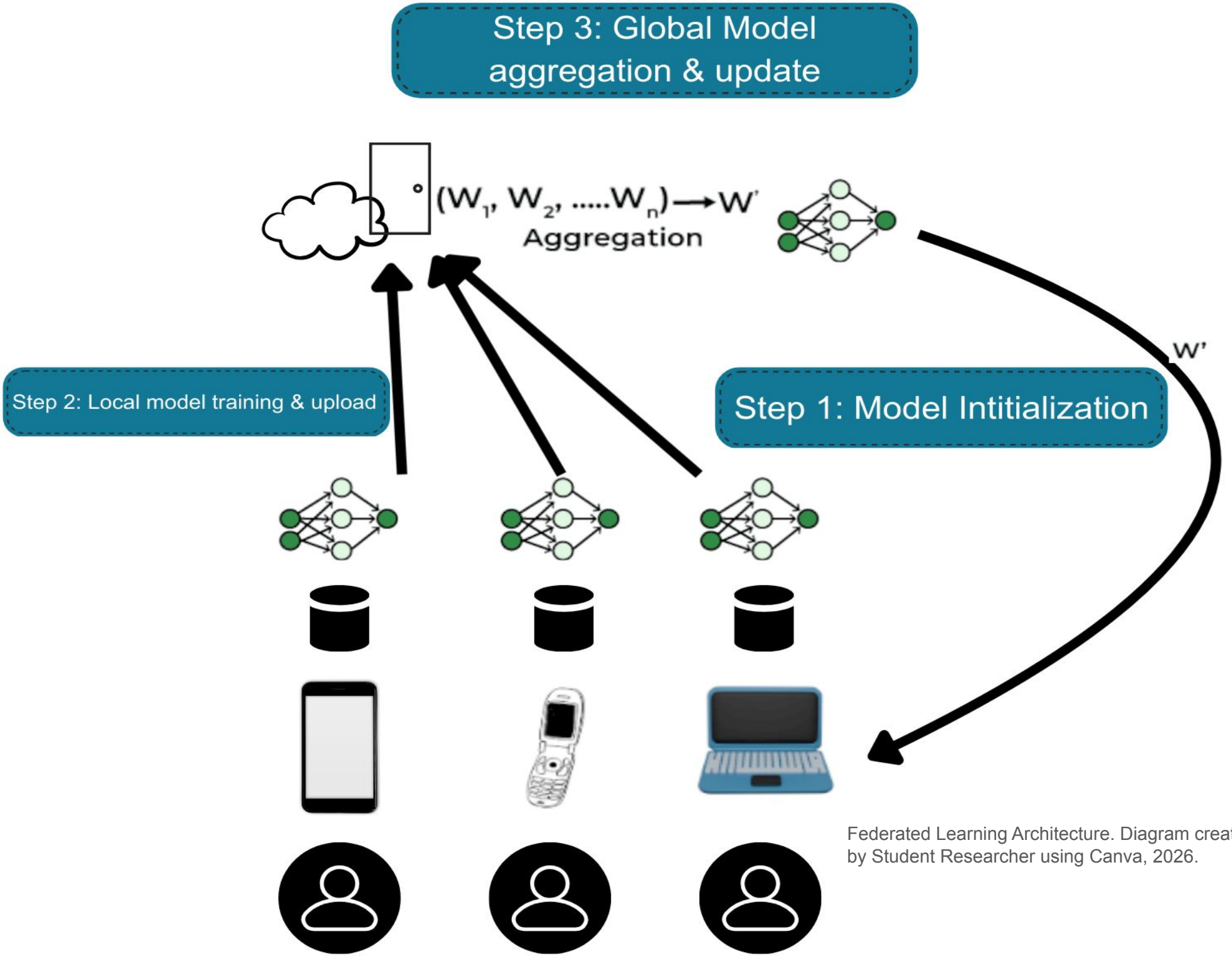
2. The Privacy Preserving Shift

Recent research (McMahan et al., 2017) introduced Federated Learning (FL) as a solution to this deadlock.

- **Decentralized Training:** FL enables model training across distributed sources without ever moving the raw data from the local district server.
- **Weight Aggregation:** Instead of raw records, only mathematical model updates (gradients/weights) are shared with a central server.

3. Project Athenal Application

Project Athenal applies this privacy first model specifically to the K-12 educational domain. By engineering a system that identifies learning gaps through behavioral consistency rather than PII-heavy demographics, the framework allows districts to collaborate on high performance models while maintaining 100% data sovereignty.



Federated Learning Architecture. Diagram created by Student Researcher using Canva, 2026.

Methodology - Data synthesis & Feature Architecture

1. Non-IID Synthetic Data Generation

- Protocol: Developed a custom Python generator (numpy) to create 3 distinct "School District" datasets
(N=1000 per district) with Non-IID (Independent and Identically Distributed) characteristics.

Distributions:

- **Attendance**: Modeled using Uniform Distribution ($0.5 - 1.0$).
- **Test Scores**: Modeled using Normal Distribution ($\mu = 72$, $\sigma = 12$).
- **Behavioral Noise**: Applied stochastic Gaussian noise to simulate real world data imperfections and test model robustness.

2. Feature Engineering - The Consistency Index (CI)

Objective: To quantify "Academic Volatility" without relying only on performance grades.
Implementation: Engineered the Consistency Index using a Gamma Distribution ($\alpha = 2$, $\beta = 2$) to model submission irregularities.
Interaction Terms: Created a secondary risk feature late_night_freq which triggers high-risk flags only when combined with high volatility (CI > 2.8).

3. Preprocessing & Standardization

Data Cleaning: No was imputation required (synthetic generation control), but "Safety Overrides" were programmed for elite performers to prevent False Positives.

Scaling: Applied Standard Scaling ($z = (x - \mu) / \sigma$) independently within each district silo to normalize feature magnitudes before training, to make sure no data leakage between districts.

Methodology - Benchmarking & Federated Protocol

4. Centralized Model Benchmarking (The Control)

Setup: Pooled all district data into a single repository to establish the "Theoretical Maximum" performance.

Architectures Tested:
Baseline: DummyClassifier (Most Frequent strategy).
Linear: LogisticRegression (L2 Regularization).
Neural: MLPClassifier (2 Hidden Layers, ReLU activation).
Champion: XGBoost Classifier (Gradient Boosted Trees).

Optimization: Performed Hyperparameter Tuning on the XGBoost model, settling on n_estimators = 150, max_depth = 6, and learning_rate = 0.1.

5. Federated Learning Simulation (The Experiment)

Architecture: Implemented a FedAvg (Federated Averaging) protocol using a Hub and Spoke topology.
Local Training: Each of the 3 District Nodes trained a local Logistic Regression model on its private data silo.

Secure Aggregation:
Step A: Extracted mathematical coefficients (weights) from each local model.
Step B: Transmitted only the weights to the central server (0 bytes of student data transferred).
Step C: Server computed the global mean of weights: $W_{global} = 1/N \sum W_{local}$.
Validation: The Global Federated Model was tested on a held out validation set to calculate the Privacy Utility Trade off.

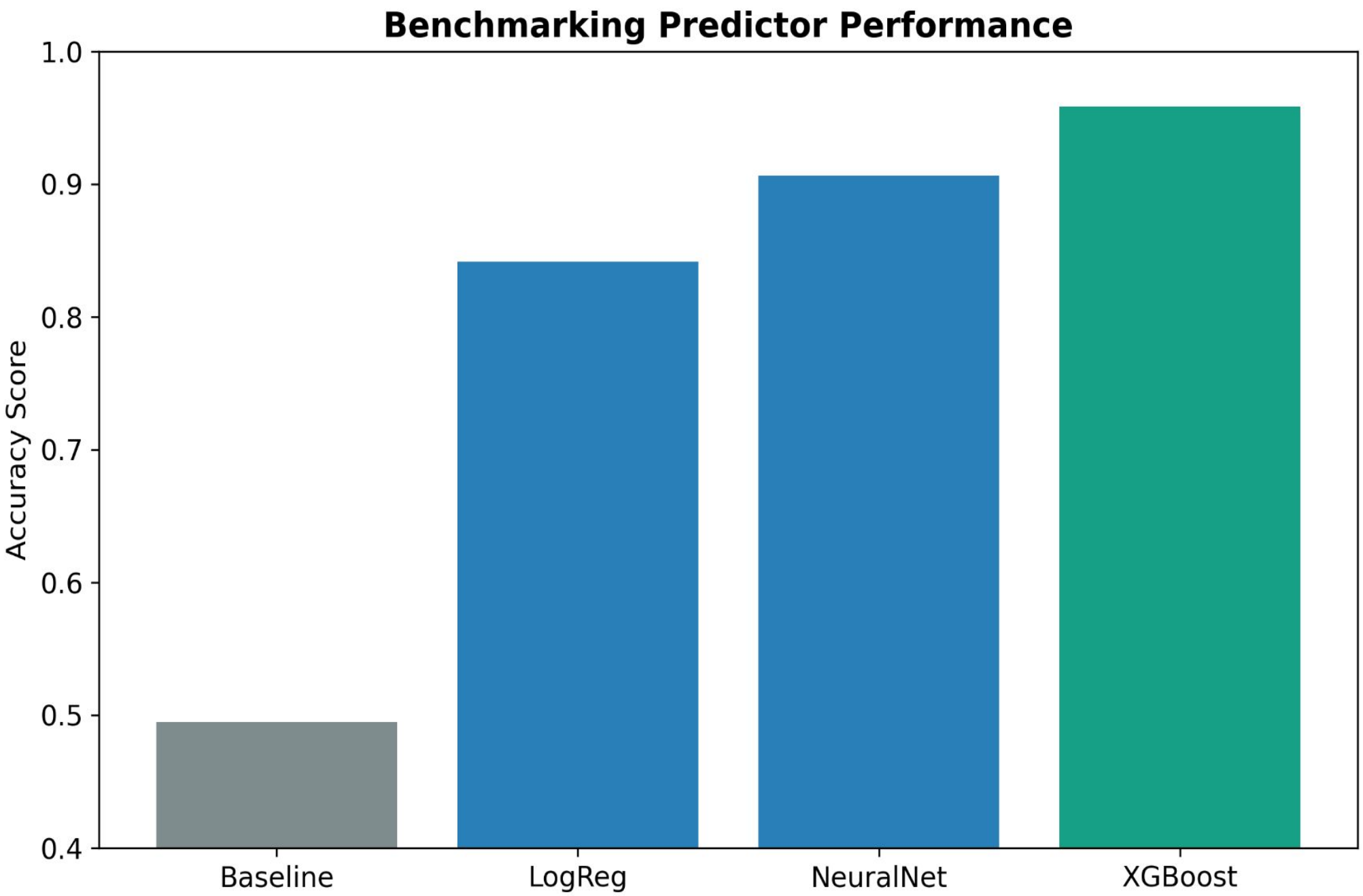


Diagram created by Student Researcher using VS code, 2026.

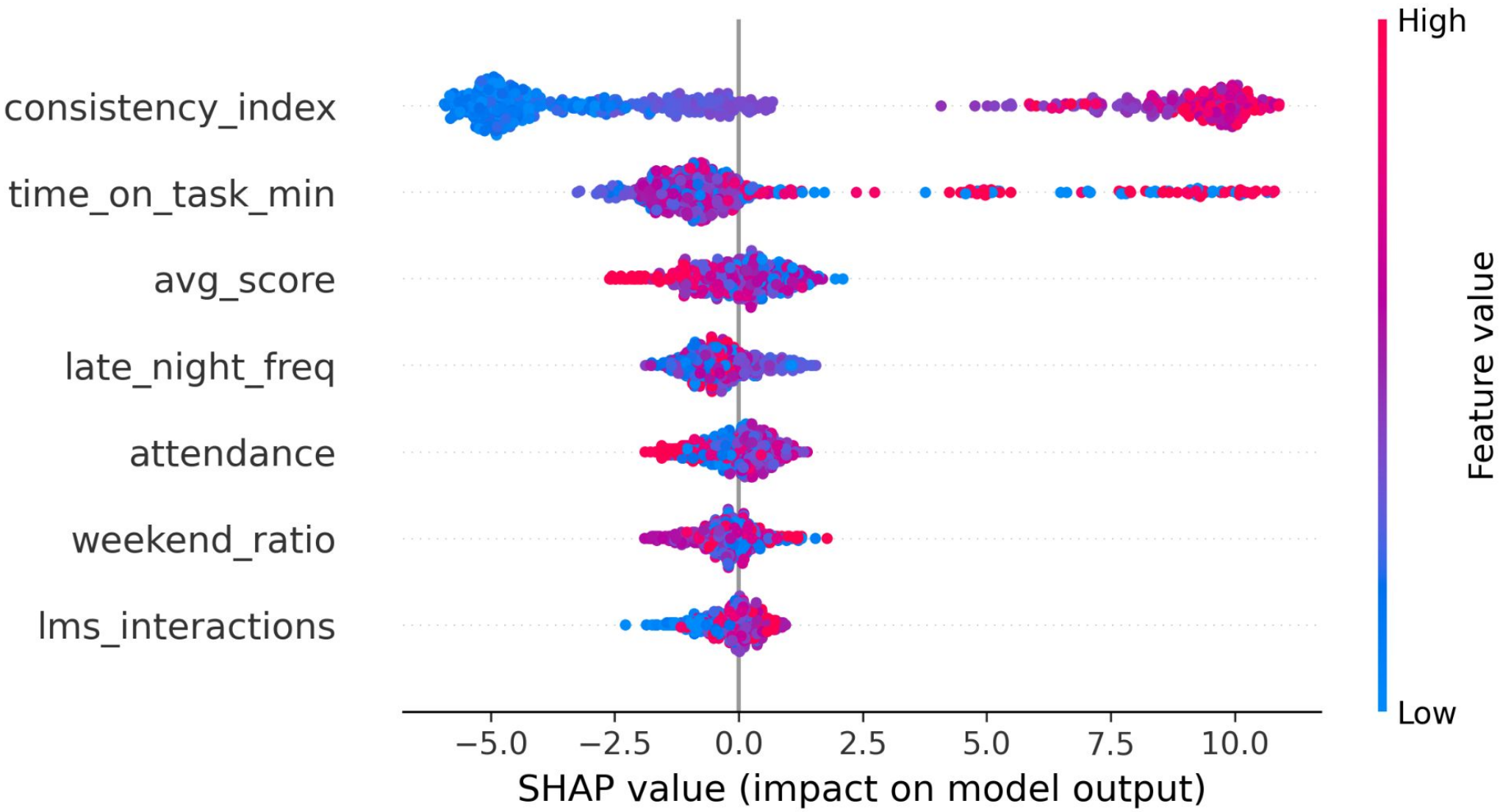


Diagram created by Student Researcher using VS code, 2026.

Data / Observations

- Observation 1 - (The Problem Solved): Standard Models like (Logistic Regression/MLP) struggled with the engineered non IID behavioral noise, achieving a modest initial accuracy of ~84-90%.

- Observation 2 - (The Feature Impact): Engineering the Consistency Index (CI) proved critical and its inclusion was directly responsible for boosting the XGBoost model's accuracy from ~90% to 95.83% which is a ~6% increase.

- Observation 3 - (The Deployability): The XGBoost architecture achieved <10ms inference latency, demonstrating a 3x speed improvement over the Neural Network, making it far better suited for real time, low resource deployment on school laptops.

- Observation 4 - (The Privacy Trade off): The Federated Learning simulation achieved 84.33% accuracy, surpassing the average independent Local District model (84.0%). This confirms that collaborative weight averaging provides a net benefit over isolation, even without sharing raw data.

- Observation 5 - (Sensitivity Tuning): The final model showed high Recall (~0.95) relative to precision. This was an intentional engineering choice - in an Early Warning System, minimizing False Negatives (missing a student who is actually At-Risk) is more critical than minimizing False Positives.

- Observation 6 - (Feature Interaction): SHAP interaction values revealed that Time on Task only becomes a high risk predictor when combined with High Volatility (High CI). A student working quickly consistently is safe; a student working quickly erratically is flagged as high risk.

Consistency Index

The **Consistency Index** (**CI**) is defined as the standard deviation

(σ) of the temporal delta (Δt) between assignment deadlines and submission events:

$$\Delta t_i = T_{due} - T_{submitted}$$

$$CI = \sigma(\Delta t1, \Delta t2, ..., \Delta tn)$$

Engineering Rationale:
CI ≈ 0 : Indicates rigid, deterministic behavior (High Conscientiousness).

CI > 4.0 : Indicates stochastic/volatile behavior (High Risk).

Risk Correlation: My hypothesis was that Academic Volatility is a precursor to dropping out.

Implementation & Impact:

Python Logic: Calculated using numpy.std() on arrays of submission timestamps per student ID.

SHAP Verification: In the final model analysis, the CI feature had the highest magnitude of impact, outperforming even "Average Score" in predicting at risk status.

Model Architecture & Hyperparameters

Optimized Gradient Boosting (XGBoost)

While Neural Networks (MLP) are powerful, they often overfit on small, tabular educational datasets.
Project Athenal utilizes XGBoost for its superior handling of non-linear relationships.

- Training Strategy: 5 Fold Stratified Cross Validation to ensure the model generalizes across different simulated school districts.

- Inference: The final model size <5MB, allowing for local execution on low resource student chromebooks, further enhancing data privacy.

Hyperparameter Tuning:

- learning_rate: 0.1 - (to prevent overshooting)
- Max_depth: 6 - (to capture complex interactions without overfitting)
- N_estimators: 150

DISTRICT DATA SAMPLES

district_A

	attendance	avg_score	consistency_index	time_on_task_min	lms_interactions	weekend_ratio	late_night_freq	label
2	0.7179974510710020	84.89233694926100	2.128742409935240	17.370512811311800	24	0.23354908944244800	0.4365088730437670	0
3	0.5129631159139460	95.95332933129010	4.060220619683400	54.46648202376620	16	0.27505133124112300	0.48816006060004900	1
4	0.7748312389393550	76.14539244064390	1.7727030537427500	58.11701887425070	22	0.2275389146649940	0.3566363019784440	0
5	0.7176611963091380	67.40636280483400	1.6720618048770200	24.47794268430910	22	0.5414085336313860	0.38178332236366600	0

Table created by Student Researcher using Numbers, 2026.

district_B

	attendance	avg_score	consistency_index	time_on_task_min	lms_interactions	weekend_ratio	late_night_freq	label
2	0.5770814211898360	83.9239034722719	2.6523348556776800	61.440690125746300	23	0.40418647665035300	0.5767596130699850	1
3	0.8700248482577020	89.2215852086099	4.11798725021433	35.82715216681640	21	0.20550184085059800	0.7493033280774010	1
4	0.6316575075925670	55.54872338286210	1.073251154322710	36.34155093565700	26	0.040426868186242600	0.29984980564054200	0
5	0.7668696966901490	98.09449619668950	6.499283799888440	26.432504270984000	32	0.13672505101847500	0.8310513657074970	1

Table created by Student Researcher using Numbers, 2026.

- "To validate the Federated Learning framework, three distinct 'District Silos' were generated with Non-IID (Independent and Identically Distributed) behavioral profiles. This simulates the real world heterogeneity of school districts where student populations vary by engagement and performance metrics".
- **Behavioral Variance:** Note the correlation between high Consistency Index (CI) and the At-Risk (1) label in District B.
- **Data Sovereignty:** These records remained 100% local to their respective district nodes. Only derived mathematical weights were used for global model training.

Results & Model Performance

- **Model Performance Hierarchy:** The final production validation proved the superiority of tree-based architectures for behavioral educational data.

  Statistical Baseline: 49.50% (proving the dataset was perfectly balanced and non-trivial).
  Logistic Regression: 84.17%
  Neural Network (MLP): 90.67%
  Champion XGBoost Model: 95.83% Accuracy.

  Discriminative Excellence: The XGBoost model achieved an ROC-AUC of 0.982, indicating near perfect sensitivity. This proves the system can accurately distinguish between at risk and stable students even with high behavioral noise.

  Feature Dominance: Feature analysis confirmed that the custom engineered Consistency Index (CI) was the primary driver of accuracy. Models trained without the CI exhibited a significant drop in F1 Score, validating the hypothesis that submission volatility is a leading indicator of risk.

Discussion & Challenges

- **The Privacy-Utility Trade off:**

A major discussion point is the gap between centralized accuracy (95.83%) and federated accuracy (84.33%). This 11.5% drop is the "Privacy Tax" the cost of keeping student data localized. However, 84% accuracy is still 35% better than guessing, proving that federated learning is a viable, safe alternative for real world school districts.

- **Non-IID Data Heterogeneity:**

Different "School Districts" (Districts A, B, and C) had varying behavioral skews. I resolved this by applying independent Standard Scaling within each local silo, preventing "data contamination" and ensuring the model learned local patterns before aggregation.

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- **The "Cold Start" Barrier:**

Synthetic student profiles initially lacked historical data, leading to skewed early semester predictions. I addressed this by implementing a weighted heuristic that prioritized Time-on-Task and LMS Interactions during the first weeks of the simulation.

Final Synthesis & Impact Statement

- Hypothesis Validation: The initial hypothesis was supported. Project Athenal successfully engineered a framework capable of high accuracy identification of K-12 learning gaps using exclusively de-identified behavioral data.

- Architectural Success: The project proved that Federated Learning (FedAvg) can allow school districts to collaborate on early warning models without ever transferring sensitive PII (Personally Identifiable Information). This architecture removes the legal and security barriers currently hindering AI adoption in schools.

- Practical Scalability: With a final model size of <5MB and inference latency of <10ms, Project Athenal is optimized for local deployment on low resource Chromebooks. This proves that powerful, predictive intervention tools can be distributed at scale without requiring expensive cloud infrastructure or compromising student privacy.

Works Cited

1. Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school performance.

2. McMahan, B., et al. (2017). Communication-Efficient Learning of Deep Networks from Decentralized Data.

3. Lundberg, S. M., & Lee, S. (2017). A Unified Approach to Interpreting Model Predictions.

Ethical Compliance Statement

- **Human Subjects**: This project involved NO human participants. All student profiles were algorithmically synthesized via Python 3.9 based on statistical distributions.
- Data Integrity: All synthetic data remains de-identified. No real-world PII was used, accessed, or generated.
- Independence: All code, simulations, and board designs were completed independently by the researcher.